

## *The low-methionine content of vegan diets may make methionine restriction feasible as a life extension strategy.*

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Recent studies confirm that dietary methionine restriction increases both mean and maximal lifespan in rats and mice, achieving "aging retardant" effects very similar to those of caloric restriction, including a suppression of mitochondrial superoxide generation. Although voluntary caloric restriction is never likely to gain much popularity as a pro-longevity strategy for humans, it may be more feasible to achieve moderate methionine restriction, in light of the fact that vegan diets tend to be relatively low in this amino acid. Plant proteins - especially those derived from legumes or nuts - tend to be lower in methionine than animal proteins. Furthermore, the total protein content of vegan diets, as a function of calorie content, tends to be lower than that of omnivore diets, and plant protein has somewhat lower bioavailability than animal protein. Whole-food vegan diets that moderate bean and soy intake, while including ample amounts of fruit and wine or beer, can be quite low in methionine, while supplying abundant nutrition for health (assuming concurrent B12 supplementation). Furthermore, low-fat vegan diets, coupled with exercise training, can be expected to promote longevity by decreasing systemic levels of insulin and free IGF-I; the latter effect would be amplified by methionine restriction - though it is not clear whether IGF-I down-regulation is the sole basis for the impact of low-methionine diets on longevity in rodents.